

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: _____

Criteria	Relevance to Application (1.25)	Use of Cloud Concepts (1.25)	Practical Insight (1)	Completeness (0.75)	Clarity (0.75)	TOTAL (5)
Weightage	25%	25%	20%	15%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Application Based Questions

1. How does cloud federation improve resource sharing and availability across datacenters?
2. What are the key security issues in virtualization, and how can they be mitigated?
3. How does a software-defined datacenter enhance scalability and management?
4. How can identity and access control ensure data privacy in cloud-based healthcare systems?
5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

1. How does cloud federation improve resource sharing and availability across datacenters?

Introduction

Cloud federation is the integration of multiple cloud datacenters or cloud providers to share computing resources such as servers, storage, networks, and applications. It enables organizations to access resources from different cloud environments as a single virtual infrastructure. Cloud federation improves scalability, reliability, and resource utilization while reducing operational costs.

How Cloud Federation Improves Resource Sharing

1. Resource Pooling

Cloud federation combines resources from multiple datacenters into a common resource pool. Organizations can dynamically allocate CPU, memory, storage, and network resources based on demand.

2. Load Balancing

User requests are distributed across multiple datacenters, preventing any single datacenter from becoming overloaded and ensuring better application performance.

3. Elastic Scalability

When one datacenter reaches its capacity, workloads can be automatically transferred to another datacenter with available resources. This ensures uninterrupted service during peak demand.

4. Disaster Recovery

If one datacenter experiences hardware failure, power outage, or natural disaster, another federated datacenter can immediately take over operations, ensuring business continuity.

5. Better Resource Utilization

Idle resources in one datacenter can be used by another, reducing wastage and improving infrastructure efficiency.

Improving Availability

- Provides redundancy through multiple datacenters.
- Ensures high availability even during failures.
- Supports automatic failover mechanisms.
- Reduces service downtime.
- Improves reliability of cloud applications.

Advantages

- High availability
- Efficient resource utilization
- Cost reduction
- Improved scalability
- Better disaster recovery
- Increased flexibility

- Enhanced performance

Conclusion

Cloud federation enhances resource sharing by connecting multiple cloud datacenters into a unified environment. It improves scalability, availability, disaster recovery, and efficient utilization of cloud resources while ensuring reliable service delivery.

2. What are the key security issues in virtualization, and how can they be mitigated?

Introduction

Virtualization allows multiple virtual machines (VMs) to run on a single physical server using a hypervisor. Although virtualization improves resource utilization and flexibility, it introduces several security challenges that must be addressed to protect cloud environments.

Key Security Issues

1. Hypervisor Attacks

If attackers compromise the hypervisor, they may gain control over all virtual machines running on the host.

2. VM Escape

A malicious program inside one virtual machine may attempt to escape the virtual environment and access the host operating system or other VMs.

3. Data Leakage

Improper isolation between virtual machines may expose sensitive data belonging to different users.

4. VM Sprawl

Uncontrolled creation of virtual machines increases security risks and management complexity.

5. Insider Threats

Administrators with excessive privileges may misuse access to confidential data.

6. Malware Attacks

Virtual machines may become infected with viruses, ransomware, or spyware if not properly secured.

Mitigation Techniques

- Use secure and updated hypervisors.
- Implement Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).
- Encrypt data at rest and during transmission.
- Regularly update and patch systems.
- Deploy firewalls and IDS/IPS.

- Monitor virtual machines continuously.
- Perform regular backups.
- Remove unused virtual machines.
- Conduct periodic security audits.

Benefits

- Improved data protection
- Stronger access control
- Reduced cyber threats
- Better compliance
- Secure cloud infrastructure

Conclusion

Virtualization security can be strengthened through proper isolation, secure hypervisors, encryption, access control, regular monitoring, and timely updates, ensuring safe and reliable cloud operations.

3. How does a Software-Defined Datacenter (SDDC) enhance scalability and management?

Introduction

A Software-Defined Datacenter (SDDC) is a modern cloud architecture where computing, storage, networking, and security resources are fully virtualized and managed through software. It enables centralized control, automation, and efficient resource utilization.

How SDDC Enhances Scalability

1. Resource Virtualization

All physical resources are virtualized, allowing them to be allocated dynamically based on workload requirements.

2. Automatic Scaling

Resources such as CPU, memory, and storage can automatically increase or decrease according to application demand.

3. Load Balancing

Traffic is distributed evenly across servers to improve application performance and prevent overload.

4. Rapid Resource Provisioning

New virtual machines and storage can be deployed within minutes without purchasing additional hardware.

How SDDC Improves Management

Centralized Management

Administrators manage all infrastructure from a single software platform.

Automation

Routine tasks such as provisioning, monitoring, and backups are automated.

Policy-Based Management

Resources are allocated according to predefined organizational policies.

Continuous Monitoring

Real-time monitoring helps identify performance issues and security threats quickly.

Advantages

- Improved scalability
- Better resource utilization
- Reduced operational costs
- Faster deployment
- Simplified management
- Increased flexibility
- Enhanced disaster recovery

Conclusion

Software-Defined Datacenters improve scalability and management through virtualization, automation, centralized control, and intelligent resource allocation, making cloud infrastructure more efficient and reliable.

4. How can identity and access control ensure data privacy in cloud-based healthcare systems?

Introduction

Cloud-based healthcare systems store sensitive patient information such as medical records, laboratory reports, prescriptions, and billing information. Identity and Access Management (IAM) ensures that only authorized users can access confidential healthcare data, protecting patient privacy and maintaining compliance with healthcare regulations.

Identity Management

Identity management creates and manages digital identities for doctors, nurses, patients, and administrators. Each user receives unique login credentials and permissions according to their role.

Access Control Mechanisms

Role-Based Access Control (RBAC)

Users receive permissions based on their responsibilities. For example:

- Doctors can view and update patient records.
- Nurses can access treatment information.
- Patients can view only their own records.

Multi-Factor Authentication (MFA)

Users verify their identity using multiple authentication methods such as passwords, OTPs, fingerprints, or biometric authentication.

Encryption

Patient data is encrypted while stored and during transmission to prevent unauthorized access.

Audit Logging

All user activities are recorded to detect suspicious behaviour and support compliance audits.

Least Privilege Principle

Users receive only the minimum permissions necessary to perform their duties.

Benefits

- Protects patient privacy
- Prevents unauthorized access
- Supports legal compliance
- Improves accountability
- Enhances trust in healthcare services

Conclusion

Identity and access control protect sensitive healthcare information by combining authentication, authorization, encryption, audit logging, and least privilege principles, ensuring secure and privacy-preserving healthcare services.

5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Introduction

Virtualization enables multiple virtual machines (VMs) to run on a single physical server using a hypervisor. This technology allows cloud providers to maximize hardware utilization, reduce costs, and efficiently allocate resources based on application demands.

Role of Virtualization

1. Server Consolidation

Multiple virtual machines share a single physical server, reducing hardware costs and power consumption.

2. Resource Sharing

CPU, memory, storage, and networking resources are dynamically allocated among virtual machines according to workload requirements.

3. Rapid Deployment

New virtual machines can be created quickly without installing new physical hardware.

4. High Availability

Virtual machines can be migrated between servers without significant downtime, ensuring continuous service.

Role of Hypervisors

Resource Allocation

Hypervisors distribute CPU, memory, storage, and network resources efficiently among virtual machines.

Isolation

Each virtual machine operates independently, preventing failures or attacks from affecting other virtual machines.

Load Balancing

Hypervisors help distribute workloads evenly across physical servers, improving system performance.

Live Migration

Virtual machines can be moved from one physical host to another while running, minimizing downtime during maintenance.

Advantages

- Improved scalability
- Better hardware utilization
- Lower infrastructure costs
- High availability
- Faster deployment
- Simplified management
- Better disaster recovery
- Enhanced resource efficiency

Conclusion

Virtualization and hypervisors are essential technologies in cloud computing. They improve scalability by enabling dynamic resource allocation and support resource efficiency through server consolidation, isolation, load balancing, and live migration, resulting in high-performance and cost-effective cloud applications.